Microseconds Are Not the Problem: A Lock-Free Soft-Real-Time Data Plane for Nanopore Adaptive Sampling, and the Signal-to-Seed Bottleneck That Bounds It

Khushal Singh
M.Tech thesis work

Abstract

Oxford Nanopore adaptive sampling ("Read Until") requires a client to decide, within a physical deadline, whether to eject a molecule still transiting the pore. We make two contributions. First, we show the decision budget is not the obstacle: a single-process C++20 data plane with single-copy ingestion, zero-copy propagation, lock-free single-producer single-consumer rings and a reference-counting arena protocol reaches a median per-chunk decision latency of 124.8 µs [115.6–132.3] against a derived service bound of 1.56 ms, at 3.70 GiB peak resident set over a 64 Mb reference index of 63.9 M minimizers, sustaining 604–645 chunk/s with zero drops across nine 40 s runs. For calibration, the bare orchestration layer of a basecalling client (readfish with a no-operation caller) costs 124.3 µs per read-chunk on the same rig—indistinguishable from our *complete* signal-space decision. Second, we show where the real limit lies. We frame Read Until as a soft-real-time *loss system*, in which a mean comfortably inside budget is consistent with an arbitrary deadline-miss rate, and we report a distribution against a derived budget rather than a mean. Our pipeline resolves 17.6% [16.6–18.8] of reads at a zero-false-positive threshold where RawHash2 resolves 91.8% [91.2–92.4] on identical input. We account for this gap without appeal to the decision rule. We proposed a seed-survival law $P(\text{key}) = a^E$ in the per-event bucket agreement a and key length E, argued it was a lower bound, and then *refuted our own bound*: once event retention is data-dependent the governing form is $P(\text{key}) \approx (1-g)^E a^E$, where g is the rate at which the two event sequences fail to correspond at all. Measured under one estimator on identical input at realistic motor-dwell dispersion, RawHash2 achieves $a = 0.911$, $g = 0.092$ at $E = 8$, giving $P(\text{key}) = 0.254$ and 467 anchors per read, against our $a = 0.783$, $g = 0.196$ at $E = 12$, giving 0.0055 and 14.9 anchors. The 46-fold key-match difference decomposes into 9.3× from encoding and segmentation and 4.9× from key length; anchor supply differs 31-fold. We identify three root causes—asymmetric event retention between reference and query encoders, a quantiser that spends too little alphabet on agreement, and recoverable loss inside the chainer—and argue that a and g should be measured and reported before any decision rule is proposed.

Keywords

nanopore sequencing, adaptive sampling, soft real-time systems, lock-free data structures, raw signal mapping, quantisation

1 Introduction

In nanopore sequencing, a molecule is read while it passes through a pore, and the instrument records the resulting ionic current in real time. Since the data arrives while the molecule is still inside the pore, the instrument offers an interface, Read Until, through which a client can ask for an unwanted molecule to be ejected, freeing the channel for another. The benefit is straightforward: on-target yield goes up, and library preparation stays the same.

This interface comes with a deadline, and the deadline is set by physics rather than by convention. Signal is delivered in chunks at a fixed rate, and each chunk describes bases that have *already* gone through the pore. If a decision arrives after the molecule has left the pore, it is not merely late. It has no value at all. Suppose C channels are active, each delivering a chunk every T_c seconds, and a decision becomes worthless once W chunks have passed. The mean service time we can allow per decision is then

$$S < \frac{W \cdot T_c}{C}$$

which at $W = 2$, $C = 512$ and $T_c = 0.4$ s gives 1.5625 ms, hereafter 1.56 ms.

Two ways of framing the problem follow from this, and we organise the paper around them.

Soft real time. A decision does not lose value gradually. Its usefulness drops to zero at the deadline. What matters is therefore the *distribution* of decision latency, not its mean.

A loss system, not a delay system. Under saturation a client must choose between queueing work and discarding it. Queueing converts decisions that would have been useful into decisions that arrive too late, so dropping is correct rather than merely pragmatic. The consequence for evaluation is sharp and, we argue, under-appreciated: **a mean comfortably inside the budget is consistent with an arbitrary deadline-miss rate.** A system reporting "mean 124 µs against a 1.56 ms budget" has said nothing about whether it meets its deadline.

Existing Read Until work does report latency, in two forms: signal consumed before a decision (mean bases or chunks per read), and mean processing time per read. What we did not find in the systems we surveyed is a deadline-miss rate against a stated service bound, a latency distribution with tail percentiles, or the fraction of decisions arriving too late to act on. That gap—reporting a distribution against a derived budget, with the instrument and environment characterised well enough for the tail to carry meaning—motivates the first half of this paper.

Contributions.

(1) A lock-free soft-real-time data plane for signal-space adaptive sampling, with a reference-counting arena protocol that makes reset-under-dispatch unrepresentable by construction rather than by a timing assumption (§3), evaluated as a distribution against a derived budget over nine runs (§4).

(2) A negative result, explained rather than excused: our pipeline is 5.2× less accurate than a published signal-space mapper on identical input, and we locate the cause before seeding (§5).
(3) A seed-survival law, including the refutation of our own earlier formulation of it, and an estimator-symmetric measurement of its two parameters for both systems (§6).
(4) Three measured root causes and the design prescriptions that follow (§7).

What this paper does not claim. We do not claim competitive mapping accuracy; we report 17.6% against 91.8% and explain it. We do not claim a novel matching method; quantised-event hashing is established [5, 6]. We do not claim trustworthy tail latency: every measurement here was taken under a hypervisor, and §9 scopes what that permits. All signal is simulated.

2 Background and related work

The interface. MinKNOW [1] exposes a bidirectional streaming RPC delivering raw signal chunks per channel and accepting actions. Read identity is carried by a string; a numeric read index was removed from the API in version 6.0, which constrains client-side data structures.

Basecalling clients. readfish [2] and similar clients basecall each chunk and map the resulting sequence. This is accurate and reuses mature tooling, but places a neural network and a sequence aligner [14] inside the decision loop.

Signal-space mappers avoid basecalling by encoding the reference into expected signal under a pore model and matching quantised query events directly. UNCALLED [3] uses an FM-index over events; Sigmap [4] and RawHash [5] use seed-and-chain over quantised event k-mers; RawHash2 [6] adds an adaptive quantiser; RawAlign [7] adds alignment to seeding; Sigmoni [8] classifies with an r-index over a small picoamp alphabet. A recent benchmarking framework, RawBench [9], decomposes raw signal analysis into reference encoding, segmentation, and matching, and evaluates three segmentation methods—t-test changepoint detection, basecaller move tables, and a learned segmenter. We adopt that decomposition, splitting matching into quantisation, seeding and chaining, because our argument needs that granularity.

The pipeline is ordered, and earlier stages bound later ones. Reference encoding → segmentation → quantisation → seeding → chaining. No chaining strategy can recover evidence that quantisation destroyed. This observation is the spine of §5–§7.

Streaming decision structures. Misra–Gries [12] and Space-Saving [13] maintain approximate frequent items in bounded space. We use a set-associative variant whose update cost is $O(\text{ways})$ rather than $O(\text{counters})$, which matters on a decision path.

Simulation. Squigulator [10] generates signal with configurable dwell-time dispersion; its default is a coefficient of variation of approximately 0.44. Icarust [11] serves a live Read Until API but emits a fixed sample count per k-mer, so it produces the CV = 0 case exactly; we patched it to draw lognormal dwell.

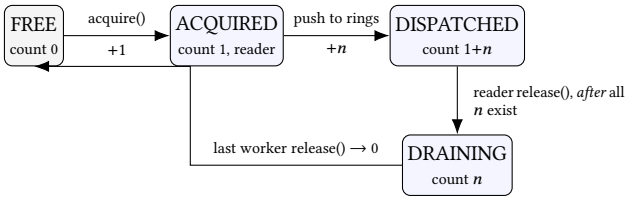


Figure 1: Arena slot lifetime. The reader holding the initial reference is what makes reset-under-dispatch unrepresentable: the count cannot reach zero while a descriptor is still in a ring, with no assumption about worker duration.

3 The data plane

3.1 Design constraints

Four constraints shape the design. The decision path must not allocate, because an allocator excursion is unbounded. It must not take a lock, because a blocked worker holds a deadline it cannot meet. State must be $O(\text{channels})$ with compile-time per-channel constants, because read length is set by library preparation and is not observable in advance. And backpressure must discard rather than queue, for the loss-system reason of §1.

3.2 Single-copy ingestion, zero-copy propagation

Signal is copied exactly once, by the protobuf runtime, into arena-backed storage at the transport boundary. Nothing downstream copies signal again. Workers receive a 56-byte borrowed descriptor holding an absolute sample index, borrowed pointers into arena-owned read-identifier and payload storage, the owning arena slot, a receive timestamp, and a narrow tail of lengths and type codes. Declared in natural reading order the structure is 72 bytes; field-ordered for size it is 56, and a compile-time assertion that it does not straddle a cacheline is what caught the difference.

Two narrowings are guarded at ingest. Chunk length drops from 64 to 32 bits because a chunk is sub-second and 2^{32} samples is about twelve days at 4 kHz. The read-identifier length drops to 16 bits because identifiers are 36-character UUIDs. The absolute sample index is deliberately *not* narrowed: a 72-hour run at 4 kHz already reaches $\approx 10^9$, uncomfortably close to 2^{32}.

3.3 The arena reference-counting protocol

The hazard is reset-under-dispatch: an arena slot recycled while a descriptor referring to it is still in a ring. Because the memory remains mapped and arena-owned, the failure is silent payload corruption rather than a crash.

The protocol removes it structurally (Figure 1). `acquire()` returns a slot at refcount 1, and that initial reference is held by the **reader**, not by a consumer. The reader pushes descriptors to n shard rings, raising the count to $1 + n$, and only then releases its own. The count can therefore reach zero only after every worker reference has been taken and dropped, and the slot cannot be reset while any dispatch is outstanding. No bound on how long a worker takes is required.

3.4 Lock-free handoff and sharding

Handoff uses single-producer single-consumer rings with free-running 64-bit counters and power-of-two capacity, so the index is a mask rather than a modulus and the full capacity is usable rather than capacity − 1. Producer and consumer each cache the other's index, removing cross-core loads from the unsaturated steady state; head, tail and both cached copies are separately cacheline-aligned. Shard rings hold 2048 descriptors; action rings hold 1024. At four shards the rings total under 0.75 MB, negligible beside the index.

Chunks are sharded by `channel & mask`, so all per-channel state is reached by exactly one thread **by construction**. This is what makes the lock-free claim cheap: there is no shared mutable per-channel state to protect. Backpressure is drop-never-block, with the invariant *offered* = *pushed* + *dropped* maintained by counters.

3.5 Bounded per-channel state

The decision structure dominates per-channel state. The set-associative vote table is $64 \times 4 \times 12$ B = 3,072 B per channel (1.57 MB at 512 channels, 8.22 MB at 2675); its update cost is $O(4)$ rather than the $O(256)$ a classical Misra–Gries table of the same counter count would pay on the decision path, and its expected retention threshold is $m/320$ against $m/257$. The anchor-history chainer is 1024×14 B = 14,336 B per channel. Every figure is a compile-time constant times the channel count; nothing depends on read length or elapsed run time.

3.6 Thread topology

Threads are placed from the kernel's SMT sibling sets rather than from processor-index arithmetic. This is not fussiness: our measurement host enumerates siblings adjacently (`0-1 2-3 4-5 6-7`), so an earlier rule that gave each role "the next logical processor" placed the reader and writer on *the same physical core* while two whole cores sat idle. The relationship is not portable. At two shards the data plane is four threads (reader, writer, two workers), which fits four physical cores; at four shards it is six, which does not.

4 Evaluation of the data plane

4.1 Method and instrumentation

Latency is the interval from the transport read returning to the decision being reached, recorded with HdrHistogram [15] at three significant figures, bounding relative error at 0.1%. The recorder reports its own backend on every line, so a degraded estimate cannot be mistaken for a measurement.

This instrumentation is a correction. An earlier power-of-two histogram reported four 99th percentiles of 111.93, 112.12, 223.66 and 447.59 µs across four runs. At the host's calibrated invariant TSC frequency of 2418.7 MHz the only values representable in that range are 54.19, 108.38, 216.76 and 433.53 µs. Every reported value was a bucket boundary; two runs "agreeing" at 112 µs agreed only that the true value lay in [54.19, 108.38]. The general form is worth stating: *any* instrument whose quantisation is coarse relative to the between-run variance will report stable values and thereby invite an inference of stability.

The harness records per run the virtualisation status, SMT topology, TSC flags, clocksource, kernel command line, governor, turbo state, free storage and load average, and refuses to label a run publishable unless virtualisation is absent, cores are isolated and the governor is set to performance. Every run reported here carries the development-only label.

Table 1: Regime R1, 37 kb reference, four repetitions per cell. Median across runs, µs.

configuration	p50	p90	p99	p99.9
pinned, 2 shards	16.3	24.7	49.1	158.0
unpinned, 2 shards	15.8	25.1	50.5	165.7
pinned, 4 shards	18.7	31.1	70.4	312.7
unpinned, 4 shards	18.4	31.6	83.8	386.7

Table 2: Regime R3, nine runs. Median across runs, with observed range, µs.

percentile	median [range]
p50	124.8 [115.6–132.3]
p90	192.8 [183.0–306.0]
p99	370.2 [346.8–1969.3]
p99.9	1224.5 [1044.9–3284.0]
mean	138.2 [131.2–206.0]

Host. Intel Core i5-1135G7, 4 physical cores / 8 logical, under WSL2. **Regimes.** R1: a 37 kb reference under constant dwell. R3: a 64 Mb human chr20 reference under variable dwell at the frozen configuration—4 bits × 12 events, minimizer [16] window 1, detected-event segmentation, two shards, pinned.

4.2 Placement and thread count

Thread count dominates placement. Both two-shard cells beat both four-shard cells at every percentile (Table 1). Six data-plane threads do not fit four physical cores.

Pinning does not reduce the median; it makes the tail reproducible. The 99th percentile across four repetitions is 46.99, 48.32, 49.97, 50.88 µs pinned—a spread of 8%—against 44.86, 45.52, 55.45, 104.45 µs unpinned, a spread of 133%. The defensible claim about processor affinity is about variance, not location.

4.3 Genome scale: regime R3

Nine 40 s runs. The daemon's startup line confirms the configuration in every log: 63,981,237 reference bases, geometry $4 \times 12 = 48$-bit keys, window 1, 63,981,218 minimizers, 32,888,127 distinct keys, 512.0 MiB table + 204.3 MiB positions.

Peak resident set is 3.70 GiB in every run—the maximum including the index-build transient; the steady-state index structures are 716 MiB. Offered chunk arrival is 604–645 per second, set by the simulator rather than the daemon. **No chunks were dropped and no deadline misses were recorded in any of the nine runs.**

At the median this is a utilisation of **0.080** against the 1.56 ms bound. A 10-chunk read completes its decisions in **1.25 ms**—a derivation, ten chunks times the measured median per-chunk latency, not an independent measurement.

The tail is where the loss-system framing earns its place. Eight of the nine runs sit strictly inside the budget through the

Table 3: Cost comparison on the same host. readfish figures are its own progress line divided out; processor and memory figures are from `/usr/bin/time -v`.

system	mode	per chunk	CPU	peak RSS
readfish	no-op caller	124.3 µs	15.6%	86.0 MiB
readfish	caller + mapper	66.2 µs	25.3%	663.4 MiB
this work	frozen, 64 Mb	124.8 µs	181.5–202.3%	3786 MiB

99.9th percentile, at 1.04–1.40 ms. The ninth does not: one run reports p99 = 1969.3 µs and p99.9 = 3284.0 µs, and is also the worst of the nine at p90 and at the mean (Figure 2).

We do not read this as a deadline miss. Across the nine runs the observed *maximum* spans 2.0 ms to 59.4 ms—two orders of magnitude in a quantity the software should not vary in—and, tellingly, the run with the largest maximum is **not** the run with the worst p99; its p99 is 630 µs. Maximum and p99 are not correlated here, which is the signature of externally injected stalls rather than of slow decisions. On a hypervisor without core isolation, this measurement cannot establish that the system meets its deadline, and cannot establish that it misses it. Stating that honestly is part of the contribution.

4.4 Comparison with existing clients

The architectural point. readfish with a no-operation caller performs no basecalling and no mapping; 124.3 µs is its orchestration alone, a lower bound on its cost. Our *complete* signal-space decision over a 64 Mb index costs 124.8 µs. The orchestration layer of a basecalling client is already as expensive as an entire signal-space decision. This does not establish superiority—a production readfish does work this configuration omits—but it does establish that the decision budget is not what stands between signal-space matching and deployment.

The cost is processor time: 181.5–202.3% of one core against readfish's 15.6%, because workers spin before yielding. We report it alongside.

Against a signal-space mapper. RawHash2, single-threaded on identical reads with the same pore model, requires 10.64 ms of processor time per read including index load, or 10.18 ms excluding the load a long-running daemon pays once. Our comparable figure is the derived 1.25 ms. The ratio is 8.2×, and must be read with §5: a latency ratio between a system that resolves reads and one that largely does not is a bound on data-plane cost, not a performance result.

5 The accuracy gap

5.1 Head-to-head

Both systems were run on the same 20 Mb chr20 reference, the same Squigulator-derived R10.4 9-mer pore model on both sides, the same SLOW5 files, three seeds, 500 reads per cell, at Squigulator's default dwell dispersion.

Our figures are true-positive rates at the lowest threshold giving zero false positives on a composition-matched shuffled control. RawHash2's are correctly-mapped fractions at its own default operating point, which is **not** zero-false-positive: 8.4–10.6% of control reads are mapped. The comparison is of magnitude, not of level, and we state it as such.

Table 4: Accuracy, three seeds, 500 reads per cell.

system	s42	s43	s44	mean [range]
this work	18.8	17.4	16.6	**17.6** [16.6–18.8]
RawHash2	92.4	91.8	91.2	**91.8** [91.2–92.4]

Table 5: Decision rules at the same operating point, three seeds.

decision rule	s42	s43	s44	mean [range]
votes (256 counters)	14.2	15.0	13.0	14.1 [13.0–15.0]
greedy, $\beta = 0$ (ctrl)	9.6	10.8	7.0	9.1 [7.0–10.8]
greedy, $\beta = 12$	17.4	16.6	15.6	16.5 [15.6–17.4]
anchor-history, $L = 48$	18.8	17.4	16.6	17.6 [16.6–18.8]
anchor-history, $L = 128$	19.6	19.4	18.0	**19.0** [18.0–19.6]
$L = 48$, occ. filter	17.8	15.8	15.4	16.3 [15.4–17.8]

A factor of 5.2 (Table 4). The rest of this paper accounts for it.

5.2 Replication is not optional at this scale

Table 5 and Figure 3 give six rules in triplicate.

The $\beta = 0$ control is the informative row. With gap tolerance removed, chaining is *worse* than the vote table—9.1% against 14.1%—as expected when it tracks 24 candidates where the table tracks 256. The improvement is therefore attributable to the tolerance, not to the change of data structure.

The best decision rule is +35% relative over voting, 19.0% against 14.1%. No rule we tried exceeded that.

Single rules vary by up to 3.8 points across seeds—the control spans 7.0 to 10.8—comparable to the gaps *between* rules, and the ranges of adjacent rules overlap. Any comparison between configurations differing by less than roughly four points requires replication. We learned this the expensive way: an earlier single-run table reported that increasing look-back from 48 to 128 did not help. It does, by 1.4 points of mean and positively in every seed; the single run that supported the old claim was a different seed from the other five rows of the same table.

Even at its best, the decision rule moves accuracy by tens of percent where the gap to be closed is a factor of five. The deficiency is not here.

6 A seed-survival law

6.1 The model, and the refutation of our own bound

Let a be the probability that reference and query assign the same quantiser bucket to corresponding events. A key packs E consecutive events, so a key matches only if all E agree, and expected anchor yield falls as a^E.

We originally argued this was a **lower** bound. Two mechanisms induce positive correlation between per-event agreements: consecutive events derive from overlapping k-mers (adjacent 9-mers share eight bases), and every event of a read is normalised by a

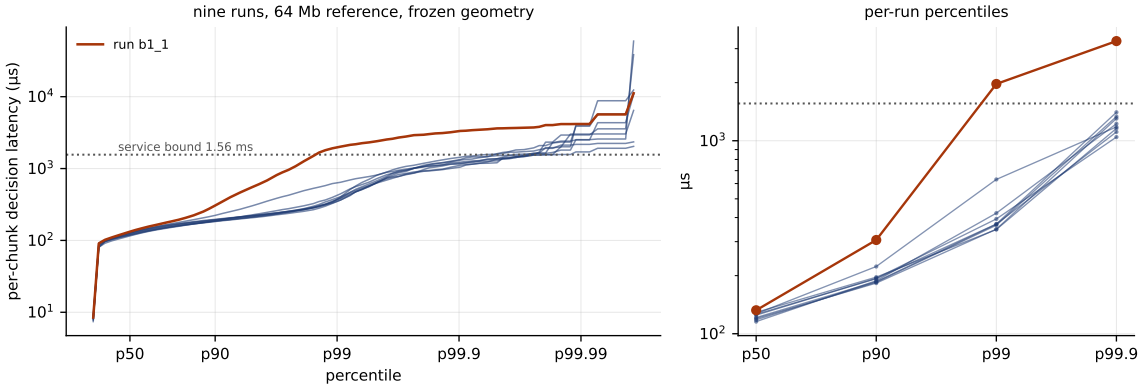


Figure 2: Regime R3 across nine runs against the 1.56 ms service bound. Eight runs lie inside the budget through p99.9; one crosses at p99 and p99.9.

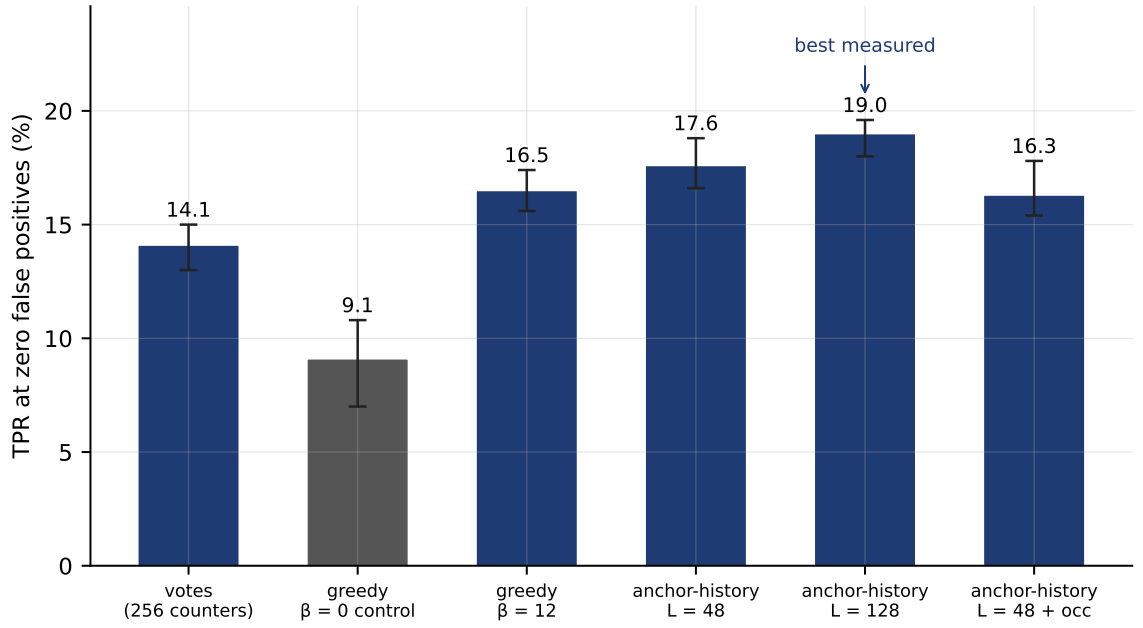


Figure 3: Decision rules with their across-seed ranges. Adjacent rules overlap, so a single run cannot separate them.

single shift-and-scale estimate, so an error in that estimate displaces all of a read's z-values together. Under positive correlation, agreements cluster, raising the probability that a run of E is entirely in agreement relative to the independent case.

That argument is conditional, and the condition fails for most configurations of interest. It holds only given an exact correspondence between the two event sequences. A key window containing a single inserted or deleted event cannot match however well its neighbours agree. Writing g for the probability that an alignment column is an indel rather than a matched pair, the governing quantity is

$$P(\text{key}) \approx (1-g)^E \cdot a^E$$

and the first factor is absent from the simple form. Where segmentation is fixed-width and dwell constant, g is small and a^E behaves as the lower bound derived above—measured at 1.8× and 5.1× below the direct count in the two such configurations. Where segmentation is data-dependent or dwell varies, g rises to 0.09–0.21 and a^E becomes an **over-estimate**, by between 1.9× and 14× across the remaining cells (Figure 6). Figure 4 shows the a^E surface over the (a, E) plane with both systems' measured operating points.

The qualitative content survives—yield falls geometrically in E, with a base that is a measurable property of the quantiser—but the use of a^E as a one-sided bound is restricted to the exact-correspondence case.

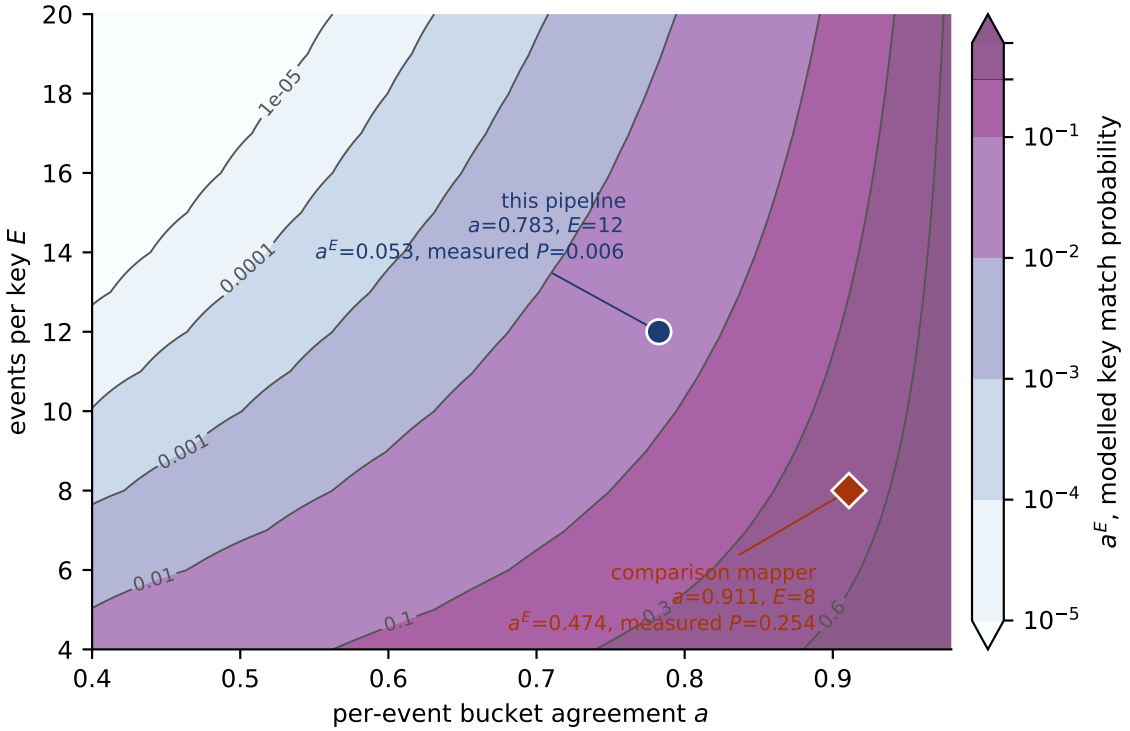


Figure 4: Modelled key-match probability over the (a, E) plane, with each system's measured operating point. Measured P sits below a^E in both cases.

6.2 An estimator-symmetric instrument

An agreement figure computed two different ways for two systems is not a comparison. We therefore emit per-event quantised symbols from both systems in one format and score them with one program, which does not know which system produced its input.

Our symbols come from the production reference-encoding and quantisation paths. RawHash2 ships `--out-quantize`, which dumps query-side symbols; its reference path hardcodes the dump off, so we added a four-line `--out-quantize-ref` exposing the same facility on the reference side, and fixed a record-separator defect that makes the stock multi-read output unparseable.

The scorer aligns the two symbol streams semi-globally—query aligned in full, free end gaps on the reference, match +1, mismatch −1, gap −1—and reports a under the maximising alignment, a under the single best constant offset, the indel fraction, and, directly from the alignment path, the fraction of positions beginning a gap-free all-matching run of E columns. That last quantity is $P(\text{key})$ **measured rather than modelled**, which removes the independence assumption entirely.

Validation. Under the conditions of our earlier per-event diagnostic, the independent instrument returns a constant-offset agreement of 0.556 against the 0.559 previously published: different code, same input, agreeing to 0.3 percentage points.

A note on estimator choice. The constant-offset estimator, which our earlier work used, understates agreement by a factor of four once retention is data-dependent: 0.188 for our frozen configuration where the correspondence-aware figure is 0.783. Results obtained with it for detected-event configurations are withdrawn, including the claim that detection roughly halves agreement. It does not.

Table 6: Agreement, correspondence failure, and measured key-match probability. 200 reads per cell, identical input, one estimator.

system	CV	a	g	E	$P(\text{key})$	anchors
this work	0.00	0.797	0.170	12	0.0124	33.8
this work	0.44	0.783	0.196	12	0.0055	14.9
RawHash2	0.00	0.913	0.055	8	0.350	651
RawHash2	0.44	0.911	0.092	8	0.254	467

6.3 Results

The gap is large and it is before seeding. At CV 0.44, measured key-match probability differs by a factor of **46** (0.254 against 0.0055) and anchor supply by a factor of **31** (467 against 14.9 per read, the two ratios differing because the systems retain different numbers of events per read; Table 6, Figure 5). A 31-fold difference in anchor supply accounts for 91.8% against 17.6% without any appeal to the decision rule.

The key-match difference decomposes, and key length is the smaller lever. Within our pipeline, measured $P(\text{key})$ at $E = 8$ rather than 12 is 0.0274 against 0.0055, a factor of **4.9**. The remaining factor of **9.3**—0.254 against 0.0274 at the same key length—is attributable to the encoding and segmentation themselves ($4.9 \times 9.3 = 45.8$, recovering the 46-fold total). Shortening keys is both the smaller lever and, as §7 shows, the one that pays for itself in lost specificity.

RawHash2's agreement is dwell-invariant where ours is not. 0.913 → 0.911 across CV 0 to 0.44, against 0.797 → 0.783 for ours, while our key-match probability falls 2.2× over the same range because g rises from 0.170 to 0.196. Agreement is not what moves; correspondence is.

7 Three root causes

7.1 Asymmetric event retention

RawHash2 uses an event-merging rule: any event whose level differs from the last retained event by less than `-sig-diff` = 0.35 is dropped. It applies this rule **symmetrically to the reference encoding and to the query**, and keeps 73.4% of events on each side. As a result, the two sides always agree on how many events a given span contains.

Our pipeline does not have this property. The reference encoder always emits one event per k-mer, whereas the query detector emits about 0.78. The effect is measurable and serious: at zero dwell dispersion, with no timing noise to blame, our indel rate is **0.170**. An indel rate that high with no noise present is a structural mismatch between two encoders, not a property of the signal. Because correspondence failure enters $P(\text{key})$ with the same exponent as disagreement, it costs as much. This is the first thing to fix, and it is a change to the indexing path rather than a research question.

7.2 A ≈2.5-bit quantisation sweet spot

RawHash2's nominal four bits admit sixteen buckets. Its realised symbol distribution over chr20 occupies six of them—7.7, 21.1, 9.7, 27.6, 24.0 and 8.8 per cent, plus a 0.9 per cent outlier bucket—for a realised entropy of approximately **2.46 bits per event**. Its quantiser is deliberately non-uniform: fine resolution inside a central band, coarse outside.

A third system points the same way, which matters because two points could be coincidence. Sigmoni [8], which classifies rather than maps and shares none of our decision machinery, quantises into an alphabet of **six** symbols by default—2.58 bits—on the stated grounds that it balances robustness to noise against specificity in exact matching. Two independent systems landing within a tenth of a bit of each other, one naming noise robustness as the reason, suggests the operating point is not accidental.

Our uniform quantiser over a clipped z-range spends its alphabet evenly, realises more of it, and agrees less. **The design axis is per-event entropy against per-event agreement, and the two systems sit at opposite ends of it.** A key of 8 events at 2.46 bits carries about 19.7 bits of nominal specificity; one of 12 events at nearly 4 realised bits carries about 48. The system with the *less* specific key is five times more accurate, because specificity it does not need costs it nothing and agreement it does need costs us everything.

Table 7: Realised key entropy and occurrence-cap survival, 20 Mb reference.

geometry	nominal	distinct keys	realised	short	cap
3×8	24	92,482	16.5	7.5	2.8%
4×8	32	1,519,844	20.5	11.5	31.2%
3×13	39	5,096,781	22.3	16.7	63.7%
3×14	42	7,985,629	22.9	19.1	78.8%
4×12	48	14,908,778	23.8	24.2	93.5%

Differencing is not the answer, and the reason is instructive. Encoding successive level *differences* cancels the shared per-read normalisation shift. Measured in triplicate it is much worse: 8.3 [3.6–11.8] against 17.6 for the absolute encoding, and worse in every seed. It retained one symbol per event and changed only what the symbol encoded, so E agreements were still required while each symbol became a function of two noisy levels rather than one. It attacked the right quantity by the wrong mechanism. What merging does instead is reduce E *and* raise a, which the $(1-g)^E a^E$ form rewards twice.

Specificity is also scale-dependent. A 39-bit key realises 22.3 bits, a shortfall of 16.7; the frozen 48-bit key realises 23.8, a shortfall of 24.2 (Table 7, Figure 7). **The shortfall grows faster than the nominal width**: doubling nominal key space from 24 to 48 bits buys 7.3 bits of realised entropy.

Short keys saturate the occurrence cap, and worse on larger references. A 3×8 geometry keeps only 2.8% of minimizer occurrences on the 20 Mb reference—97.2% truncated—and 0.7% on the 64 Mb one, 99.3% truncated. Short keys degrade faster than the reference grows, so a geometry chosen on a small reference does not transfer to a large one. This is why raising anchor yield by shortening keys fails: realised entropy collapses before the yield becomes useful.

7.3 Recoverable loss inside the chainer

The correspondence-aware measurement revises our frozen configuration's anchor supply from a modelled 1.85 per read—derived from the simple a^E form for a *fixed-width* configuration we do not ship—to a measured **14.9** for the configuration we do. The observed median best-chain score is **4**.

So roughly 11 of 15 anchor-eligible positions per read are not reaching the best chain. This loss was invisible while the estimate sat *below* the observation: a decision stage cannot recover more anchors than exist, and 1.85 against 4 left no room for loss. The candidates are the occurrence cap discarding anchors before chaining, the 24-slot greedy chainer thrashing, and a chaining band of 12 against a measured indel rate of 0.196. We have not isolated which.

This qualifies, but does not overturn, the location of the deficiency. Closing a 31-fold supply gap dominates recovering a factor of under four. But the decision stage is no longer blameless, and because the anchors are known to exist, this is a diagnostic rather than a search.

8 Discussion

Measure a and g before proposing a decision rule. Both are a single pass over paired reference and query events, with no training

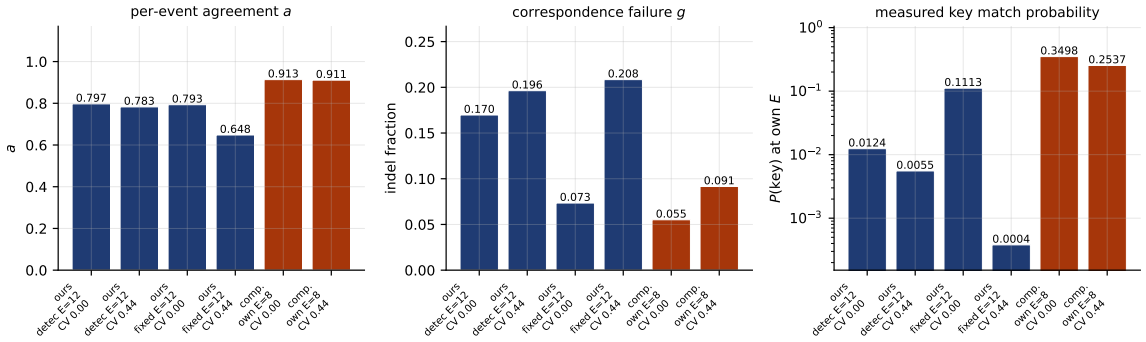


Figure 5: Both systems under one estimator on identical input.

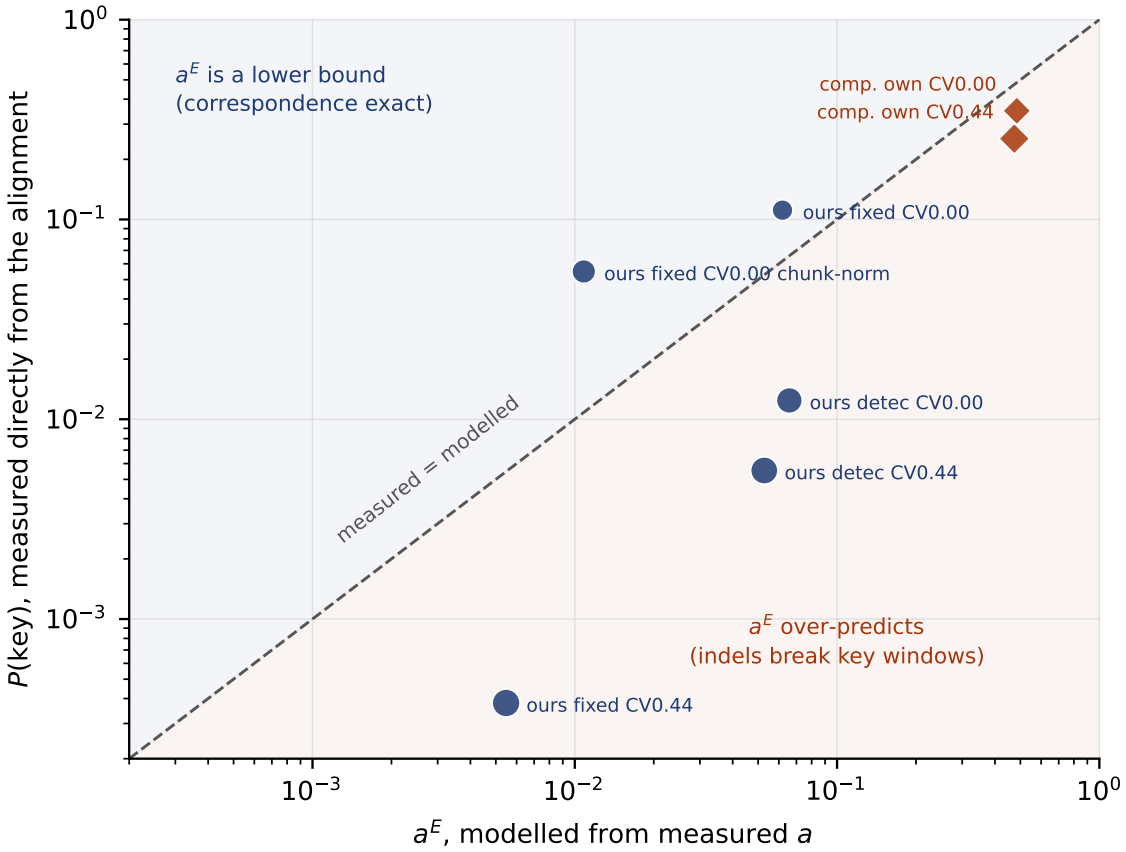


Figure 6: Measured key-match probability against the a^E prediction, with marker area scaled by the correspondence-failure rate g. a^E is a lower bound only where correspondence is exact.

and no search, and together they establish whether *any* decision rule can succeed before one is designed. A reported decision rule unaccompanied by reported agreement and correspondence figures cannot be assessed by a reader.

Optimise key-match probability, not agreement alone. Correspondence failure enters with the same exponent as disagreement, and our indel rate is 0.170 even with no timing noise present. The resource that can buy both is alphabet size. RawHash2 and Sigmoni trade it away; we do not.

Report distributions against derived budgets. Under a deadline, a mean that falls inside the budget is consistent with any miss rate. We report nine runs with full ranges and name the one that

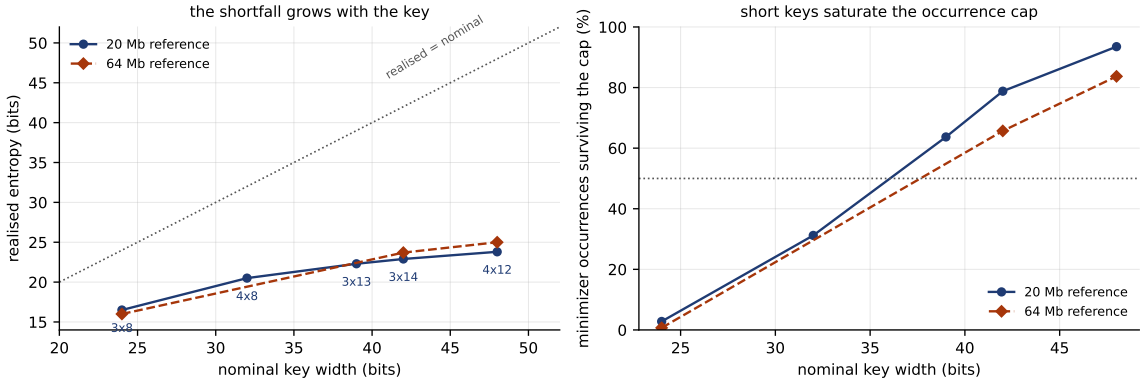


Figure 7: Realised entropy and cap survival by geometry, at both reference sizes.

crosses, which is a weaker-sounding but more useful statement than a single comfortable mean.

Run the external baseline first, not last. Every intervention we made on the decision stage was measured against our own earlier numbers, which rose and fell credibly and gave no indication that the ceiling lay elsewhere. One run of a published tool on identical input would have established within a day that the gap was a factor of five and not a tuning problem. In a pipeline whose stages are ordered, an internal improvement series measures progress against the wrong denominator until an external reference fixes the scale.

The symmetric-retention explanation is not isolated. It is visible in RawHash2's source and consistent with every measurement, but the decisive ablation—applying our detector to our own reference encoding and re-measuring—was not made.

Single reference, single organism. All figures derive from human chr20.

Related-work provenance. RawHash2, Squigulator, readfish and Icarust were built and run here; claims about UNCALLED, Sigmap, RawAlign and Sigmoni rest on their publications and repositories rather than on execution.

9 Limitations and threats to validity

Every latency figure was taken under a hypervisor. WSL2 can deschedule a vCPU mid-decision, indistinguishable in the data from a slow decision. We treat p50 and p90 as properties of the implementation and p99 and above as upper bounds contaminated by the platform. The 2.0–59.4 ms spread of observed maxima, uncorrelated with p99, is the direct evidence. A deadline-miss rate against the 1.56 ms bound requires bare metal with isolated cores, tickless operation, interrupt affinity moved off the measured cores and a fixed governor; we did not have it, and no claim here depends on the tail.

All signal is simulated, from two generators whose detector calibrations do not transfer to each other by a factor of thirty. There is no basis for assuming either transfers to a physical device.

Agreement is measured at two dwell levels, one seed, 200 reads per cell, where accuracy is swept at six levels in triplicate. The claim that RawHash2's agreement is dwell-invariant rests on two points.

The alignment-based agreement figures are upper bounds, computed under the correspondence that maximises matches. Indel fractions of 0.055–0.208 mean the alignments are near-diagonal, so the estimator is not manufacturing matches through heavy gapping, but the bound is one-sided and the comparison is only as fair as the two systems' indel fractions are similar. They are not identical, and the asymmetry favours RawHash2.

10 Conclusion

We set out to determine whether the adaptive-sampling decision can be made in microseconds. It can: a median of 124.8 µs at genome scale against a 1.56 ms physical budget, with bounded per-channel state, no allocation on the decision path, and no drops at the offered rate—while the orchestration layer alone of a standard basecalling client costs 124.3 µs per read-chunk before any decision is made. The decision budget is not the obstacle to signal-space adaptive sampling.

The question we answer better than the one we asked is why that is insufficient. Quantised-event matching is bounded not by the rate at which decisions can be made but by the fidelity with which signal survives segmentation and quantisation, captured by two measurable quantities whose E-fold product governs everything downstream. At the agreement and correspondence our pipeline achieves, no decision rule placed after quantisation can recover what segmentation and quantisation discard—which is why six attempts to place one moved accuracy by at most 35% relative against a gap of a factor of five.

That is a negative result about this system and a positive one about where to look. We state it without apology: a system that fails for a reason that is measured, modelled, and shown to account for the failure of six attempts to fix it is of more use to the next designer than one that works for reasons its author cannot name.

Reproducibility

All code, data and scripts are in the accompanying repository at commit `eb0ea28`. `scripts/verify_thesis_numbers.py` recomputes every number from files under `results/`, or, for figures describing code that no longer exists, checks them against the commit message that recorded them; it exits non-zero on any disagreement, and reports 143 claims agreeing and none disagreeing. `scripts/verify_paper_numbers.py` performs the same check against this manuscript. `scripts/generate_thesis_figures.py` regenerates every data figure from `results/` and nothing else, byte-stably. Five claims are not settled by any committed file and are documented as scope boundaries: a bare-metal deadline-miss rate, real instrument data, agreement across the full dwell sweep and across seeds, the Icarust accept rates behind the simulator comparison, and the index resident set at minimizer window 10.

References

[1] Oxford Nanopore Technologies. *MinKNOW API*. https://github.com/nanoporetech/minknow_api

[2] A. Payne, N. Holmes, T. Clarke, R. Munro, B. J. Debebe, M. Loose. Readfish enables targeted nanopore sequencing of gigabase-sized genomes. *Nature Biotechnology*, 2021.

[3] S. Kovaka, Y. Fan, B. Ni, W. Timp, M. C. Schatz. Targeted nanopore sequencing by real-time mapping of raw electrical signal with UNCALLED. *Nature Biotechnology*, 2021.

[4] H. Zhang, H. Li, C. Jain, H. Cheng, K. F. Au, H. Li, S. Aluru. Real-time mapping of nanopore raw signals. *Bioinformatics*, 37(Suppl. 1), 2021.

[5] C. Firtina, N. Mansouri Ghiasi, J. Lindegger, G. Singh, M. B. Cavlak, H. Mao, O. Mutlu. RawHash: enabling fast and accurate real-time analysis of raw nanopore signals for large genomes. *Bioinformatics*, 39(Suppl. 1), 2023.

[6] C. Firtina, M. Soysal, J. Lindegger, O. Mutlu. RawHash2: mapping raw nanopore signals using hash-based seeding and adaptive quantization. arXiv:2309.05771.

[7] J. Lindegger, C. Firtina, N. Mansouri Ghiasi, M. Sadrosadati, M. Alser, O. Mutlu. RawAlign: accurate, fast, and scalable raw nanopore signal mapping via combining seeding and alignment. arXiv:2310.05037.

[8] V. Shivakumar, O. Ahmed, S. Kovaka, M. Zakeri, B. Langmead. Sigmoni: classification of nanopore signal with a compressed pangenome index. *Bioinformatics*, 2024.

[9] M. Eris, C. Firtina, O. Mutlu et al. RawBench: a comprehensive benchmarking framework for raw nanopore signal analysis techniques. *ACM-BCB*, 2025. arXiv:2510.03629.

[10] H. Gamaarachchi et al. Squigulator: simulation of nanopore sequencing signal data. *Genome Research*, 2024.

[11] R. Munro, A. Payne, M. Loose. Icarust: a real-time simulator for nanopore adaptive sampling. https://github.com/LooseLab/Icarust

[12] J. Misra, D. Gries. Finding repeated elements. *Science of Computer Programming*, 2(2), 1982.

[13] A. Metwally, D. Agrawal, A. El Abbadi. Efficient computation of frequent and top-k elements in data streams. *ICDT*, 2005.

[14] H. Li. Minimap2: pairwise alignment for nucleotide sequences. *Bioinformatics*, 34(18), 2018.

[15] G. Tene. HdrHistogram: a high dynamic range histogram. https://github.com/HdrHistogram/HdrHistogram_c

[16] M. Roberts, W. Hayes, B. R. Hunt, S. M. Mount, J. A. Yorke. Reducing storage requirements for biological sequence comparison. *Bioinformatics*, 20(18), 2004.